Exam Date & Time: 31-Dec-2024 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru
BTech III Semester - Make up Examination - December -2024
Circuits and Systems [ECM-2123]

Marks: 50 **Duration: 180 mins.**

PART-A

Answer all the questions.

1) Calculate the equivalent resistance between A and B in the network of Fig. 1A.

A)

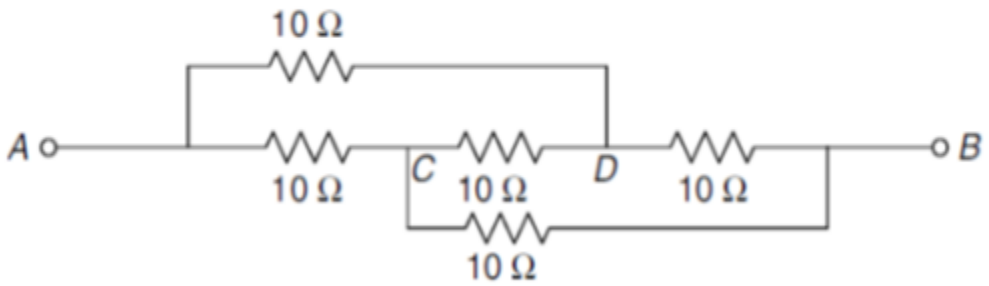


Fig: 1A

(2)

B) Obtain the Thevenin's equivalent circuit and determine the current through the 24 Ω resistor for the circuit shown in Fig. 1B.

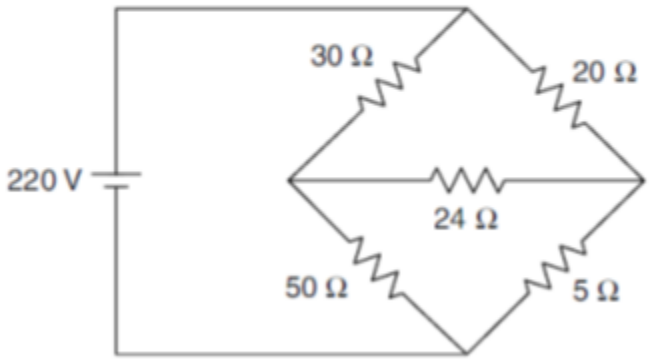


Fig: 1B

(4)

C) (4)

Using Superposition theorem, calculate the current through the 4 Ω resistor in Fig. 1C.

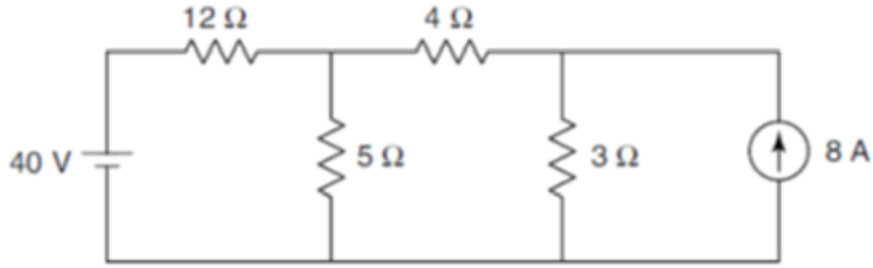


Fig: 1C

2) In the network shown in Fig. 2A, the switch is closed at t = 0. Determine the following at $t = 0^+$

A)
a. v
b. dv/dt
c. d^2v/dt^2

(5)

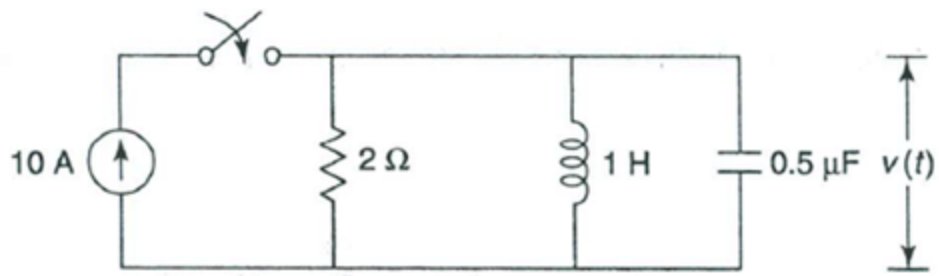


Fig: 2A

B) (5)

In the network shown in Fig. 2B, the switch K is closed at t = 0 with the capacitor initially unenergized. Obtain the particular solution for the current i(t).

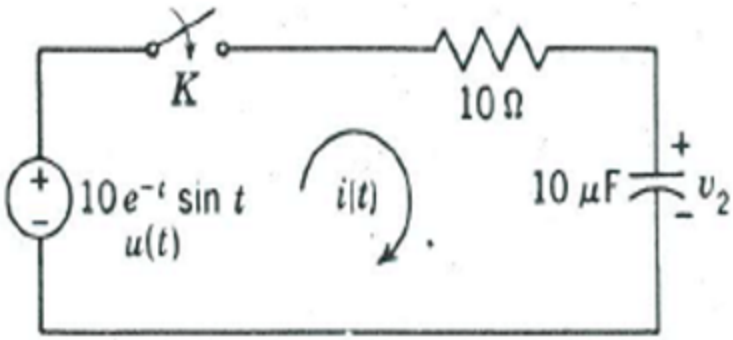
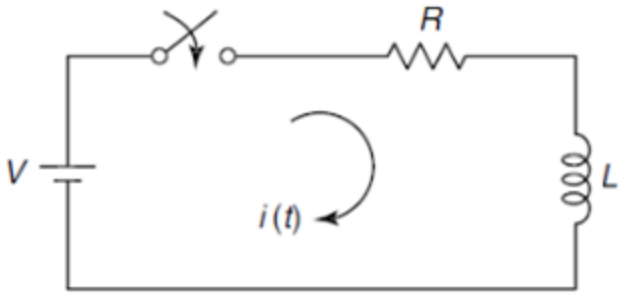


Fig: 2B

3) A general network consisting of a resistor and an inductor is shown in Fig. 3A. The switch is closed at

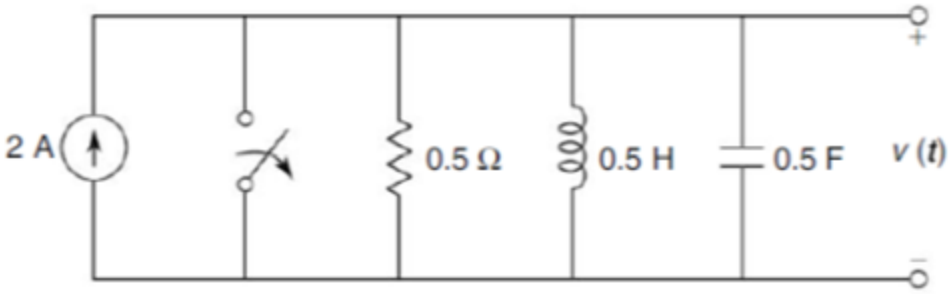
A) time t = 0. Using Laplace transform, obtain the expression for i(t) for t > 0.



(4)

Fig: 3A

B) The switch in Fig. 3B is opened at time t = 0. Using Laplace transform, determine the voltage v(t) for t > 0.



(4)

Fig: 3B

C) The network shown in Fig. 3C has acquired steady-state at t < 0 with the switch open. The switch is closed at t = 0. Determine v (t) by using Laplace transform method. (2)

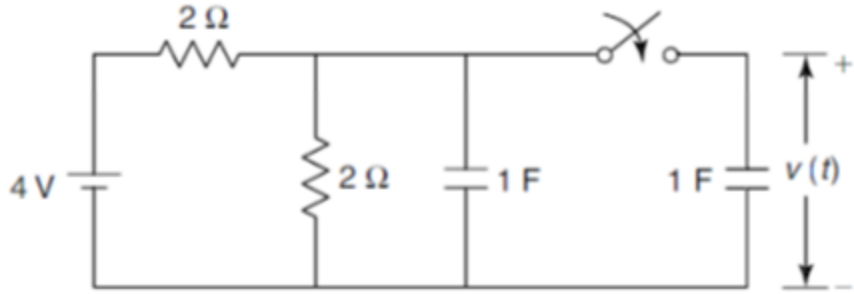


Fig: 3C

4) Sketch the following signals (4)

a. $x(t) = u(t) - u(t-2)$

b. $y(t) = u(t+1) - 2u(t) + u(t-1)$

A)

B) Determine whether the following systems are LTI systems or not, (4)

a) $y(t) = at^2 x(t) + bt\, x(t-4)$

b) $y[n] = a^n x(n)$

C) Determine if the following system is causal and memoryless (2)

$$y(t) = x(2-t) + x(t-4)$$

5) Obtain the block diagram representations for systems described by the following difference equations:

A) (4)

a) $y(t) = -\left(\frac{1}{2}\right)\frac{dy(t)}{dt} + 4x(t)$

b) $\frac{dy(t)}{dt} + 3y(t) = x(t)$

B) What are the sufficient conditions for the existence of the Fourier series of a periodic function $x(t)$? (3)

C) Obtain the Fourier transform of the signal $x(t) = e^{-2t}cos(5t)\, u(t)$ (3)

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